ATA27 Flight controls
Boeing 727



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BOEING

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2. Introduction.

The Boeing 727, a narrow-body trijet aircraft, is designed to operate on short and medium-range routes with a flight envelope ranging from approach speeds of approximately 130 kt up to altitudes of 42,000 ft. Its flight control system is based on a conventional indirect mechanical control architecture, in which the primary control surfaces (ailerons, elevator and rudder) are actuated by a network of tension cables, pulleys and push rods that mechanically transmit the pilot's inputs from the flight deck to the servoactuators of each surface, without the intervention of flight control computers.

3. General Description of the Aircraft Flight Control System

The primary flight control surfaces generate movement around X, Y, and Z axes, supported by mechanical and hydraulic systems to reduce pilot effort. The following table lists the PFC of the B727 and figure 3.1 illustrates their position.

Pitch	Unit.	Roll	Unit.	Yaw	Unit.
Elevator	2	Inboard aileron	2	Upper rudder	1
THS	1	Outboard aileron	2	Lower rudder	1
		Flight spoilers	6		

The secondary control surfaces enhance lift and enable efficient braking during takeoff and landing.The following table lists the SFC of the B727 and figure 3.1 illustrates their position.

Secondary / trim & tabs	Unit.	Trailing edge	Unit.	Leading edge	Unit.
Elevator tab	2	Inboard triple-slotted flaps	2	Leading edge flaps	4
Rudder anti-balance tabs	2	Outboard triple-slotted flaps	2	Leading edge slats	8
Aileron balance tabs	2				

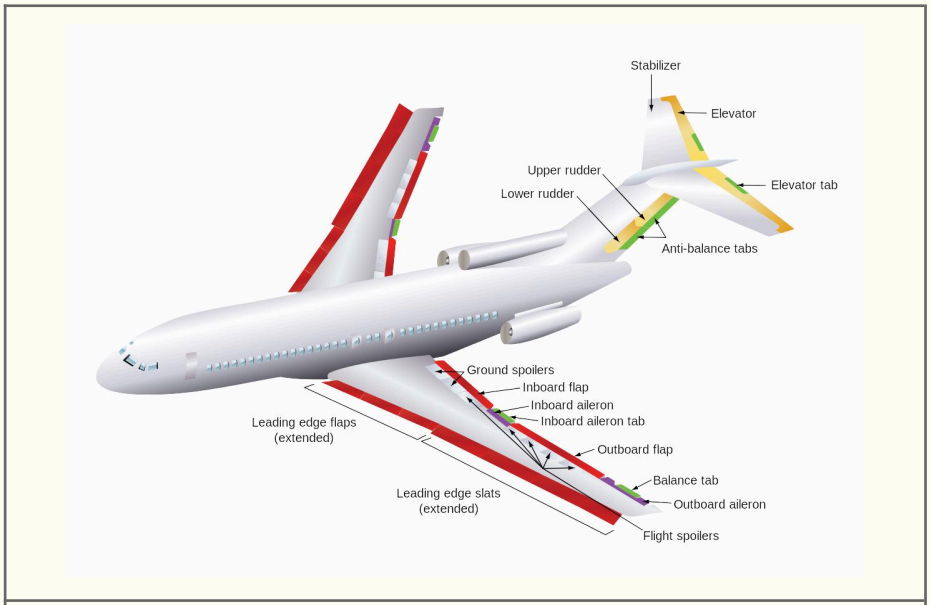


Figure 3.1

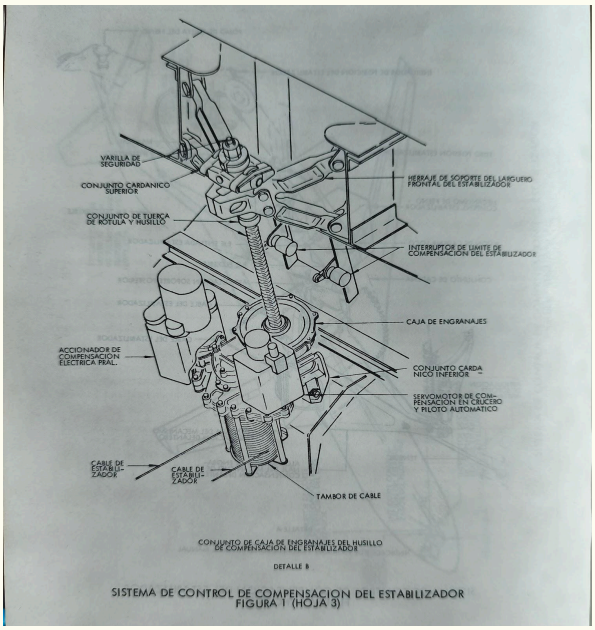
4. Operation of the Flight Control System

4.1. PITCH CONTROL

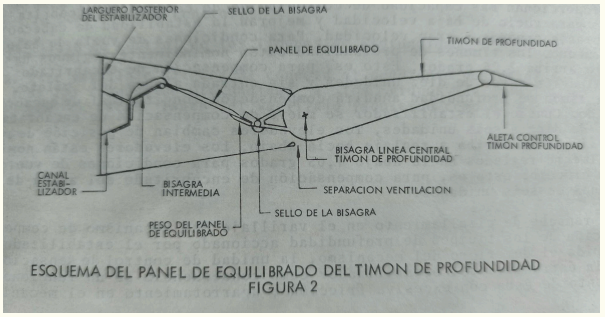
Pitch control is achieved through the **elevators**, which respond to the pilot's control column inputs (pulling back raises the nose, pushing forward drops it). Mechanical signals are transmitted to a rear sector that drives connecting rods to the **Power Control Units (PCUs)**. These units house internal valves that direct hydraulic fluid to generate the high forces required for surface deflection. To provide tactile feedback, an **artificial feel system** uses a hydraulic roller against a cam to simulate air resistance. This pressure is regulated by a computer based on airspeed (Pitot) and stabilizer position. Safety is ensured by a "Feel Diff Press" warning (triggered by a 25% pressure discrepancy between systems A and B) and a **mechanical spring package** that provides fixed centering if hydraulics fail.

In the event of a total hydraulic pressure loss, the aircraft enters **manual reversion mode**, transitioning to a purely mechanical architecture. The PCUs automatically trigger a **bypass valve** to prevent hydraulic lock, allowing the pilot’s input to actuate the **control tabs** directly. These tabs use aerodynamic force to deflect the main elevators, assisted by **balance panels** that utilize pneumatic pressure differentials within sealed cavities to minimize control column loads.

Longitudinal trimming is performed by pivoting the entire **Trimmable Horizontal Stabilizer (THS)** through a 15.3-degree arc using a **jackscrew and gearbox assembly**. This system features triple redundancy: a main electric motor, an autopilot motor, and a **manual trim wheel** on the flight deck pedestal, which is physically linked to the jackscrew via cables and chains for use during total power loss.



SISTEMA DE CONTROL DE COMPENSACION DEL ESTABILIZADOR
FIGURA 1 (HOJA 3)



ESQUEMA DEL PANEL DE EQUILIBRADO DEL TIMON DE PROFUNDIDAD
FIGURA 2

4.2. ROLL CONTROL

4.2.1. System Architecture and Command Logic

The Boeing 727's lateral control is based on a hybrid architecture that merges high-precision hydraulic actuation with the uncompromising reliability of a mechanical command chain. The pilot's input, initiated at the control wheel, is transmitted via a complex network of cables and quadrants to the Spoiler Mixer.

This "system brain" integrates aileron deflection with flight spoiler assistance. It is not merely an actuator interface but an integrated management system designed to modulate aerodynamic loads and safeguard the aircraft's structural integrity across the entire operational envelope.

4.2.2. Surface Segmentation and Wing Spar Protection

Surface management is strategically divided to optimize roll efficiency while protecting the wing spar from excessive stress. The following table outlines the operational logic of the primary surfaces:

Surface	Operational Logic	Primary Function
Inboard Ailerons	Always Active	Provides constant roll authority in all flight phases.
Outboard Ailerons	Lockout Mechanism	Neutralized when flaps are retracted to prevent torsional load at high speeds.
Flight Spoilers	Differential Deployment	Destroy lift on the descending wing to augment roll rate at low speeds.

4.2.3. Envelope Protection: The Ratio Changer

A critical element in protecting the airframe is the **Ratio Changer**. This mechanical component adjusts the system's gain dynamically based on flap configuration and airspeed.

High-Speed Configuration: The device mechanically limits spoiler travel to prevent aggressive roll maneuvers that could compromise the wing's structural integrity.

Structural Safety: By "interpreting" the aircraft's physical limitations, the system prevents the pilot from inadvertently demanding loads beyond the **VMO/MMO** design limits.

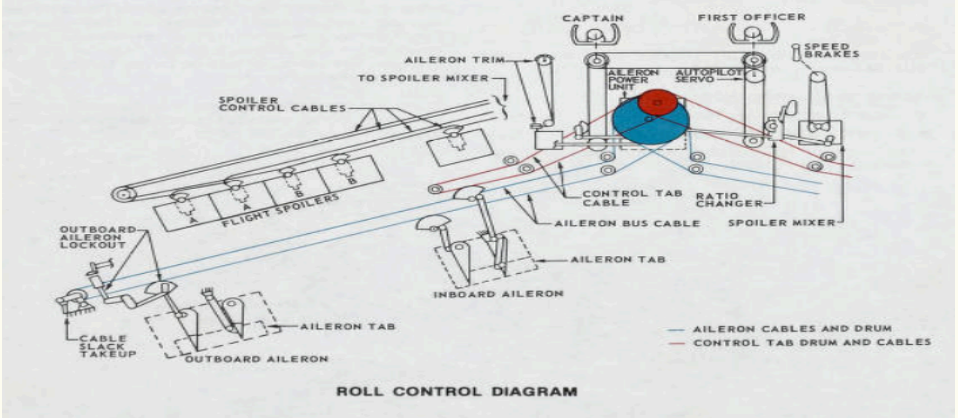
4.2.4. Redundancy and Manual Reversion

Safety is hardwired into the system through its **manual reversion capability**. In the event of a total dual hydraulic failure (Systems A and B), the control system remains functional through the following mechanisms:

Bypass Mode: Power Control Unit (PCU) valves transition to bypass, allowing direct physical control.

Servo-Tabs: These small surfaces on the aileron trailing edges become primary. As the pilot rotates the wheel, the tab moves, creating an aerodynamic force that assists in pivoting the main aileron.

This "old school" design philosophy ensures that even in a "dark cockpit" scenario, the B727 maintains effective lateral control, albeit requiring significantly higher physical effort from the flight crew.



ROLL CONTROL DIAGRAM

4.3. YAW CONTROL

4.3.1. Command Architecture and Tactile Feel

The Boeing 727 directional control system operates under a mechanical command architecture with hydraulic actuation, designed to provide precise yaw control through independent upper and lower surfaces.

- **Transmission:** Control inputs from the cockpit pedals travel via cables and linkages to the aft control quadrant located in the vertical stabilizer.
- **Feel Mechanism:** A feel and centering mechanism, utilizing a spring-loaded cam and roller system, provides a tactile response proportional to the pilot's applied effort.
- **Automatic Centering:** This configuration ensures that the rudder automatically returns to a neutral position once pedal pressure is removed, providing a consistent control reference.

4.3.2. Trim System and Equilibrium

The trim system is functionally integrated into the centering mechanism to optimize pilot workload.

- **Operation:** By using a cable drum in the cockpit pedestal, a screw mechanism is actuated to physically shift the entire feel assembly.
- **Reference Adjustment:** This adjustment changes the system's "zero" reference point.
- **Objective:** It allows the pilot to maintain a neutral rudder position against asymmetric aerodynamic forces—such as during engine failures—without constant pedal force by reconfiguring the centering spring preload.

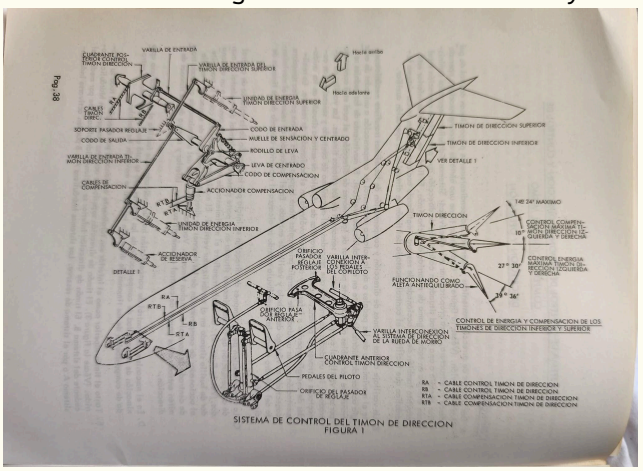
4.3.3. Hydraulic Actuation and Servo Precision

- Physical actuation of the surfaces is managed by modular hydraulic power units powered by systems A and B, incorporating a 28V DC shut-off valve for isolation and a compensator for temperature-driven volumetric fluctuations. The main servo valve directs fluid to the actuator based on rudder input, while a mechanically linked summing lever provides feedback logic. This feedback ensures the servo valve closes once the rudder reaches the demanded position, maintaining precise control.

4.3.4. Redundancy and Structural Protection

The design incorporates multiple layers of safety to ensure control continuity and structural integrity.

- Standby system dedicated to the lower rudder automatically activates if System A pressure drops below 600-675 psi. To prevent structural damage during high-speed flight, the system reduces maximum differential pressure from 2420 to 800 psi when flaps are retracted, limiting rudder deflection while maintaining full maneuverability during flap-extended phases.



5. HIGH-LIFT DEVICES

5.1. System Configuration and Kinematics

The Boeing 727 is equipped with a **triple-slotted Fowler flap system**, organized into inboard and outboard groups on each wing. This design significantly increases both the wing's camber and its surface area.
Kinematic Segments: Each flap assembly consists of three distinct segments: the **Fore Flap (Main)**, the **Mid Flap** (which rides on independent carriages), and the **Aft Flap**.
Extension Sequence: During the initial 10 degrees of travel, the segments move as a single unit to maximize wing area. Beyond 10°, a system of **sequence bellcranks and carriages** forces the segments to separate. This creates high-energy slots that allow high-pressure air from the lower surface to re-energize the boundary layer on the upper surface, preventing aerodynamic stalls at high angles of attack.

5.2. Command and Control Loop

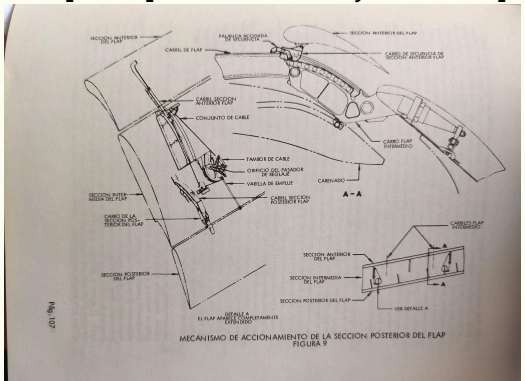
The flight crew manages the configuration via a **Flap Control Lever** with gated detents ranging from 0° to 40°.
Transmission: Movement is sent via **WFA and WFB cable systems** to the main control valves.
Control Valves: These are slide-type, **four-way hydraulic valves** that direct pressurized fluid from **Hydraulic System A** to the drive motors.
Follow-Up Mechanism: A mechanical feedback loop using a **follow-up drum and linkage** monitors the actual position. Once the flaps reach the selected angle, the linkage moves the control valve back to its **Neutral position**, hydraulically locking the system.

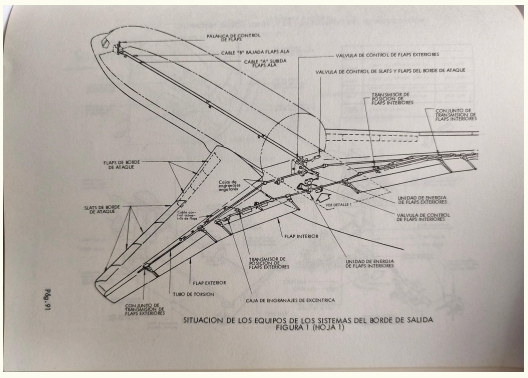
5.3. Power Generation and Mechanical Drive

The system converts hydraulic potential into mechanical torque using specialized power units:
Hydraulic Motors: Two **reversible, piston-type motors** provide the primary rotational force.
Reduction Gears: Each motor drives a power unit containing **epicyclic (planetary) reduction gears** to amplify torque.
Transmission Train: Torque is distributed through a series of **Torque Tubes**. This path utilizes **Angle Gearboxes** to navigate structural bends and **Inversion Gearboxes** to ensure symmetrical rotation between the left and right wing sections.

5.4. Actuation Mechanisms

The final movement of the surfaces is executed by high-precision mechanical actuators:
Ball Screw Assemblies: Each flap is driven by two **Ball Screw Transmission Units**. These units convert the torque tube's rotation into linear travel via a ball nut attached to the flap carriage.
Tracks and Carriages: The flaps are supported by **forged steel curved tracks**. Roller-equipped carriages ensure smooth travel along the track's aerodynamic profile.
Friction Brakes: To prevent **"blow-back"** (unintended retraction due to air loads), the transmission units include **"no-back" friction brakes** that lock the flap in position if hydraulic pressure is lost.





5.5. Leading Edge Devices (Krueger Flaps and Slats)

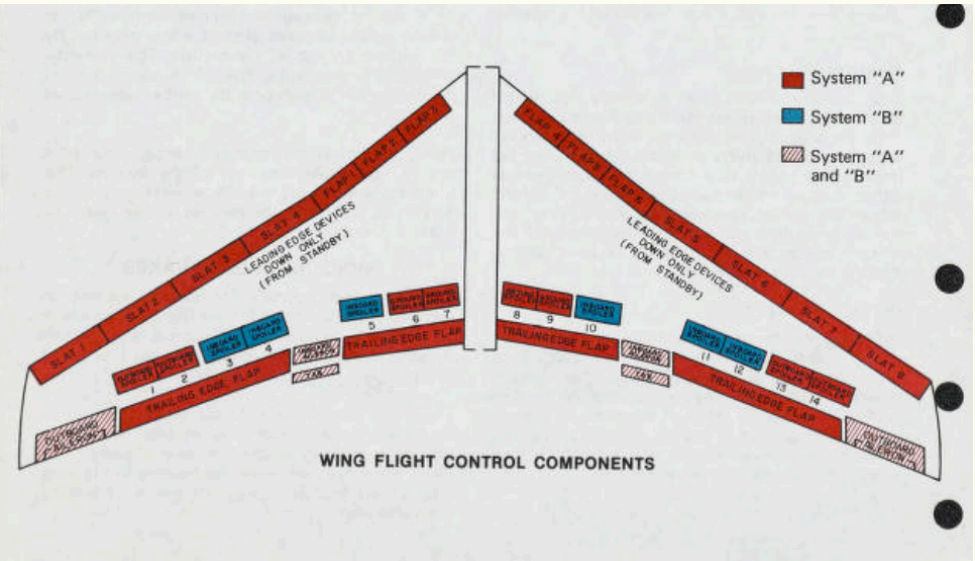
The leading edge system uses a combination of two device types to improve low-speed lift:
Krueger Flaps: Three units per wing on the inboard section, made of **magnesium** and driven by hydraulic cylinders.
Slats: Four units per wing on the outboard section, which move on guide rails and feature an **integrated thermal anti-ice system** using engine bleed air.
Deployment Sequence: Moving the lever to the **2° position** extends all Krueger flaps and slats 2, 3, 6, and 7. At the **5° position**, the remaining slats (1, 4, 5, and 8) are deployed.
Mechanical Locking: Each device uses **locking segments or keys** to secure it in the extended position, preventing aerodynamic blow-back. Hydraulic pressure is required to physically release these locks for retraction.

5.6. Protection and Monitoring Systems

Asymmetry Protection: Synchro transmitters monitor the left and right flaps. If a differential between **8° and 20°** is detected, an asymmetry detector relay triggers a **bypass valve**, halting operation to prevent roll-control issues.
Torque Limiters: These protect the wing from mechanical jams, calibrated to trip at approximately **8,370-9,370 lbs** for outboard flaps and **9,125-10,625 lbs** for inboard flaps.
Cockpit Indication: Status is monitored via a panel of annunciators: **Amber Light:** Surface is in transit. **Green Light:** Surface is fully extended and mechanically locked. **Lights Off:** Surfaces are fully retracted.

5.7. Alternate Operating Systems

In case of a total loss of System A hydraulic pressure:
Trailing Edge: An **electric 115/200V AC motor** drives the torque tubes through an alternate gearbox at a reduced speed.
Leading Edge: An **electric standby motor-pump** provides pressure to the actuators through **shuttle valves**. Note that this alternate system **only allows for extension**; there is no provision for alternate retraction.
El sistema incluye un Módulo de Prueba de Posición que permite verificar la integridad de las bombillas y el estado individual de cada superficie mediante un interruptor de prueba (posiciones FLAP POSITION, LIGHT TEST y OFF). Un interruptor de fiador en la palanca de control coordina la iluminación de acuerdo con la posición seleccionada (2° o 5°), asegurando que la tripulación sepa si la configuración real coincide con la seleccionada.



WING FLIGHT CONTROL COMPONENTS

6. LIFT-REDUCTION DEVICES

6.1. Spoiler System

The system consists of **14 total panels**: ten **flight spoilers** (five on each wing, numbered 1-5 and 10-14) and four **ground spoilers** (two on each wing, numbered 6-9).

6.1.1. Hydraulic Distribution

The actuation is purely hydraulic, distributed between two systems to ensure redundancy:

Hydraulic System A: Powers flight spoilers 1, 2, 13, 14, and all four ground spoilers (6, 7, 8, and 9).

Hydraulic System B: Powers the remaining six flight spoilers (3, 4, 5, 10, 11, and 12). Electrically operated **shut-off valves**, controlled from the overhead panel, allow the crew to cut hydraulic pressure to either system if a malfunction occurs.

6.1.2. Mechanical Control Components

The system utilizes a complex network of cables and mechanisms to translate pilot inputs:

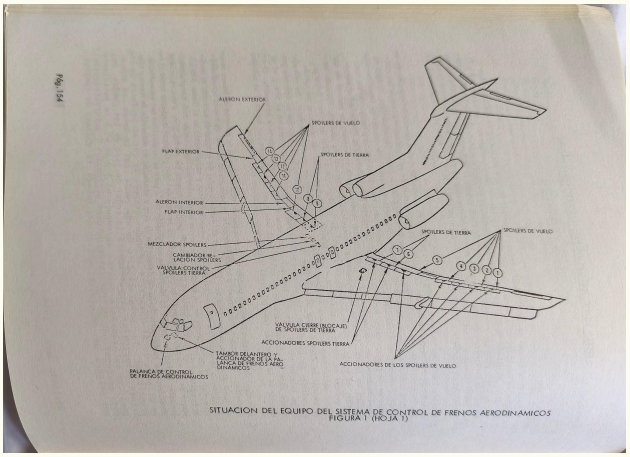
Spoiler Mixer: This is the core of the system. It combines mechanical inputs from the control wheel (lateral control/roll) and the speed brake lever. It sums these movements to position each panel individually.

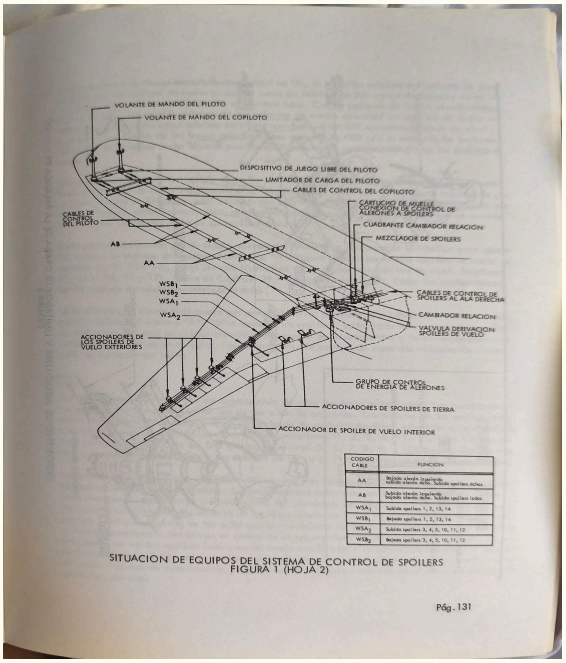
Ratio Changer: Located in the nose wheel well, this mechanism modifies spoiler response based on **flap position**. It adjusts the mixer's output magnitude to ensure smooth and effective lateral control during different phases of flight.

Lost Motion Device: Installed in the co-pilot's control column, this device ensures that if the aileron control system jams, the pilot can still operate the spoilers via the co-pilot's wheel.

6.1.3. Panel Actuators

Each spoiler has its own **hydraulic actuator**, which includes an integral control valve and a **follow-up (feedback) mechanism**. When a mechanical signal is received, the control valve directs pressure to the piston; as the panel moves, the follow-up linkage eventually recedes the valve to neutral, stopping the spoiler at the commanded angle.





6.2. Speed Brake System

The speed brake system uses the same spoiler panels but deploys them symmetrically to increase drag and reduce lift.

6.2.1. Manual Operation

Controlled by the **speed brake lever** on the center pedestal, which can be moved from 0° to 45°.

In Flight: Only the **ten flight spoilers** extend when the lever is moved. If the pilot simultaneously turns the wheel for roll control, the spoilers on one wing will rise further while those on the opposite wing will retract slightly to maintain lateral control.

On the Ground: When the lever is moved to the maximum extension position, all **14 panels** (flight and ground) deploy to approximately 45° to assist in braking.

6.2.2. Automatic Speed Brake Operation

The Boeing 727 features an automatic deployment system for landings or rejected takeoffs (RTO).

Arming Logic: For the system to operate automatically, the lever must be placed in the **ARMED** position (illuminating the green "SPEED BRAKE ARMED" light).

Activation: Deployment occurs upon touchdown when specific conditions are met: thrust levers at **IDLE** and a **wheel spin-up** signal (greater than 60 knots) or activation of the **air/ground safety sensors** (strut compression).

Automatic Retraction: If any thrust lever is advanced (e.g., during a Go-Around), the speed brake lever automatically snaps back to the "DOWN" position, retracting all panels to facilitate the climb.

6.3. Safety and Warning Mechanisms

Shear Pins: Located in the mixer and actuators, these pins are designed to break during a mechanical jam, allowing the rest of the system to continue operating or allowing a jammed panel to be retracted by aerodynamic forces.

Configuration Warning: A takeoff warning horn sounds if an attempt is made to take off with the speed brakes extended.

6.3.1. General System Description

The system is designed to allow aircraft operation on minimum-length runways by utilizing high-lift, low-weight devices. These devices consist of a combination of **Kruger-type flaps** and **leading edge slats**.

Configuration per wing: Each wing has four slats (numbered 1 through 8 from outboard to inboard) and three leading edge flaps (numbered 1 through 6 from left to right).

Construction: The slats are of conventional sheet metal construction with thermal anti-ice protection, guided by tracks and actuated by hydraulic cylinders. Kruger flaps are magnesium castings with integral ribs and stiffeners, attached by "gooseneck" type hinges.

6.4. Hydraulic Actuation System

The primary system uses **Hydraulic System "A"** for normal operation, while a **standby (alternative) hydraulic system** exists for use if the primary system fails.

6.4.1. Normal Operation

The extension of the devices is automatically coordinated with the position of the trailing edge flaps via a **spool-type control valve** operated by a follower drum.

2-Degree Position: When trailing edge flaps extend to 2 degrees, signals are sent to extend the **inboard slats (2, 3, 6, and 7)**.

5-Degree Position: Upon reaching this position, the control valve moves to extend the **remaining slats (1, 4, 5, and 8)** and all **leading edge flaps**.

7. Summary Table

System	Component	Axis	Operation & Control Logic	Hydraulic System
Primary Controls	Ailerons	Roll (X)	Wheel cables to PCUs; Outboards lock at high speed.	Sys A & B (Manual Reversion).
	Elevators	Pitch (Y)	Column cables to PCUs; Artificial feel based on speed/THS.	Sys A & B.
	Rudder	Yaw (Z)	Pedals to PCUs; Pressure reduced (800 psi) with flaps up.	Sys A (Upper) / B (Lower). (Manual Reversion)
	THS	Pitch Trim	Electric motor or manual jackscrew pivots entire structure.	Electric / Mechanical.
Secondary Controls	TE Flaps	High-lift	Lever moves hydraulic motors, torque tubes, and ball screws.	Sys A (Alt: Electric).
	Krueger Flaps	High-lift	Automatic extension at flap 5° via hydraulic cylinders.	Sys A (Backup: Standby).
	LE Slats	High-lift	Staged extension (Pos 2° and 5°) via hydraulic actuators.	Sys A (Backup: Standby).
Lift-Reduction	Flight Spoilers	Roll / Drag	Mixer combines wheel and speed brake lever inputs.	Sys A & B.
	Ground Spoilers	Lift Loss	Auto-deploy on ground; assisted by wheel spin-up sensors.	Sys A.
System Logic	Spoiler Mixer	Integration	Sums mechanical inputs to position individual panels.	Mechanical.
	Ratio Changer	Protection	Limits spoiler travel based on flap position/airspeed.	Mechanical.
Hydraulics	System A	Primary	Powered by EDPs (Eng 1 & 2); runs Gear and TE Flaps.	3,000 psi.
	System B	Secondary	Powered by EMDPs; independent of engine rotation.	3,000 psi.
	Standby System	Emergency	Dedicated pump for LE devices and lower rudder backup.	2,200 - 3,000 psi.

8. Conclusion

The Boeing 727 flight control system merges high-precision hydraulic power with a robust mechanical command chain. Safety is guaranteed through triple redundancy and manual reversion, which allows flight crews to maintain authority via aerodynamic tabs even during a total dual hydraulic failure.